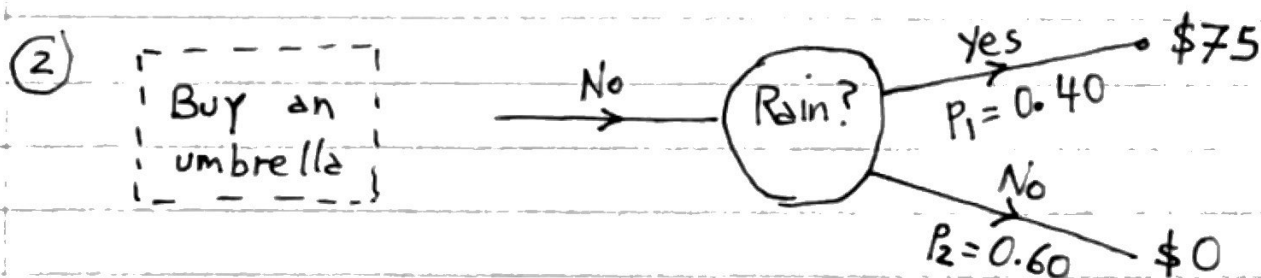


TIM 172A, MOT-I (10/16/25)  
Lecture #7

Agenda:

0. Review the 6-step DA process  
(done)
1. Back-log from L #5 : Developmental Goals
2. Aggregate Project Plan : An introduction
3. Summary of the course : Past, Present, Future
4. Reverse Engineering (aka Product Dissection)  
using FAST
5. HW # 4 : Decision for complex Product  
Development projects  
please try HW # 5 before Tuesday  
class meeting

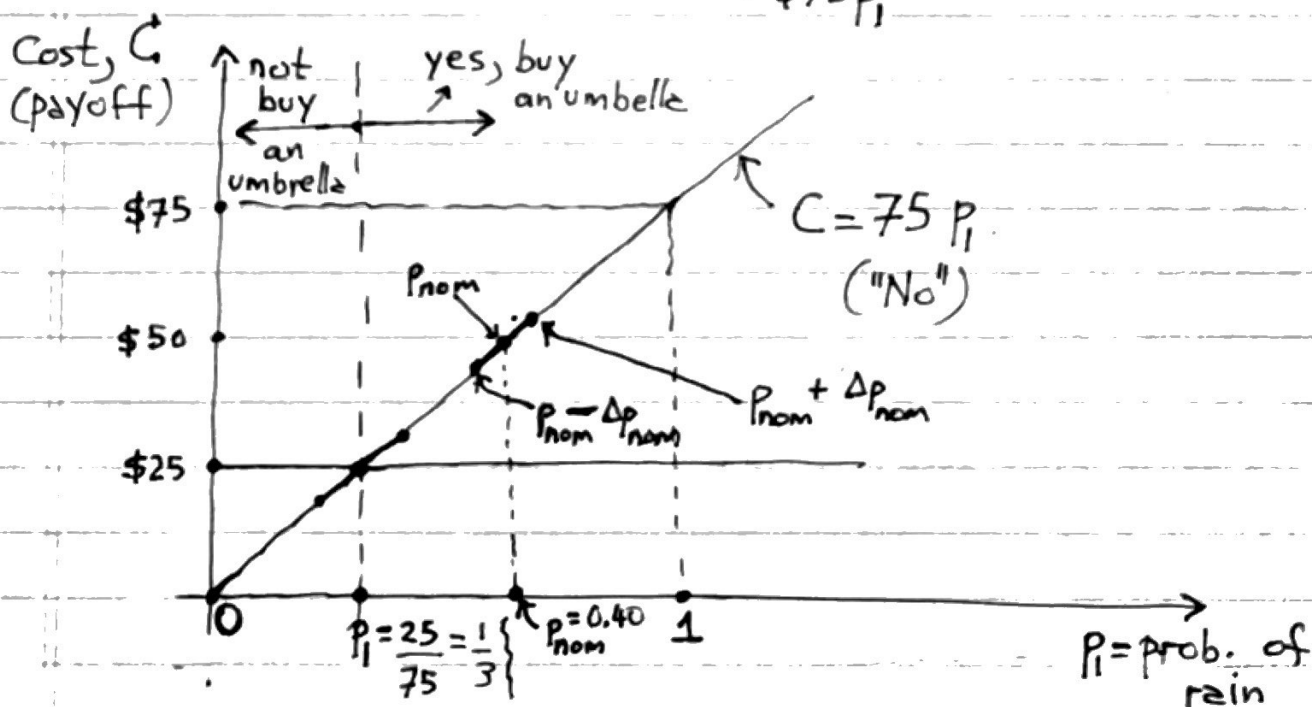
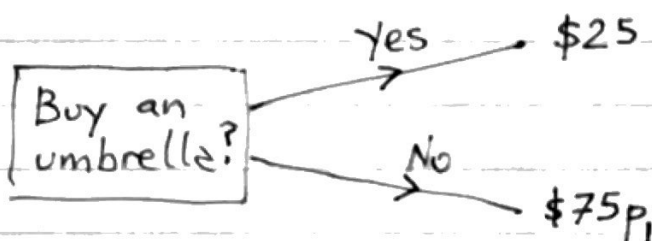


$$\text{Payoff ["Not Buy" or "No"] (cost)} = (\$75)P_1 + (\$0)P_2$$

$$= \$75 P_1$$

dependent on the probability of rain

Summary:



## 1.6 Developmental Goals (DGs) ( in L# 7, Thursday)

DGs :

- refer to concrete technology, product, and market goals
- consistent with (or aligned to) the company's competitive, technology, and P/M strategy
- in order to achieve the company's business goals (1.3)

### Examples of DGs

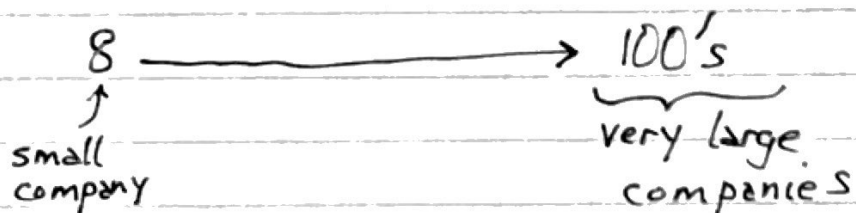
- What products & technologies should the company develop over the next year? over the next 3 years? ....
- What sales volume (# of units of the product sold per year) should the company target for the next year?
- What is the target price for each unit of the product?

∴ ∴ ∴ ∴

## 2. Aggregate Project Plan (APP)

From L #6, & the upcoming HW #4, we know how to calculate the payoff or expected monetary value (emv) for individual product development projects

In any tech. company, there will, in general be several of these projects

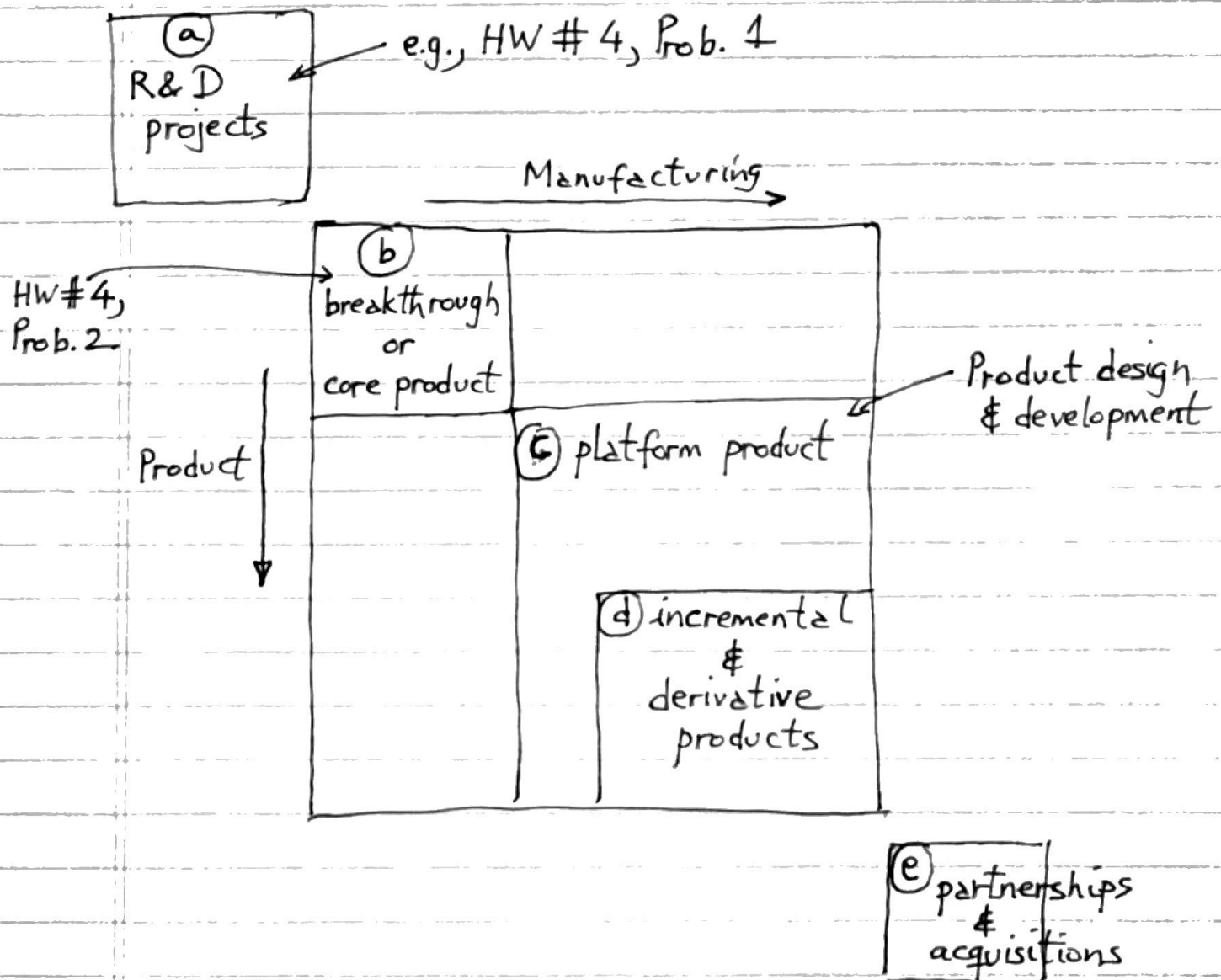


The budget for doing these projects is obviously limited.

Q1: What projects should the company work on during the next year, in order to meet its developmental goals (revenue, profit,...)?

Q2: What are the potential projects from which to decide on the actual projects to work on?

## "Potential Projects"



Also refer to Exhibit 3.8 of the text,  
U&E, PD&D, : Chapter: Product Planning

Key Question: How do we use Decision Analysis (DA) to create an Aggregate Project Plan (APP)?

see upcoming Lecture #9

### 3. Summary of the course Past, Present, & Future

Doing  
the  
right  
things!



- High-level issues ("Management")
  - Strategy (CS, TS, PMS)
  - Developmental Goals
  - Aggregate Project Plan

} USA

Doing  
things  
right



- Transition to Product Design  
& Development  
QFD/HOQ

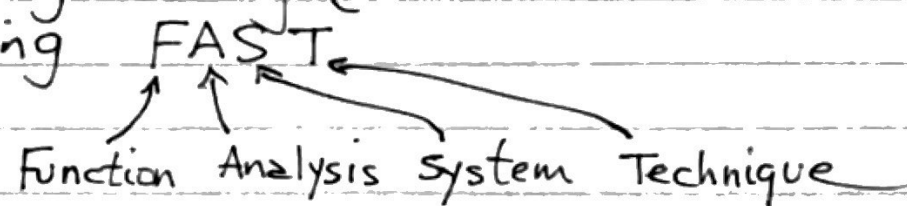
} JAPAN

- Product Design & Development

Germany

#### 4. Reverse Engineering (aka Product Dissection) using FAST

Function Analysis System Technique



##### Definitions :

- (i) Function is a verb-noun phrase that describes the purpose (or WHY?) of a system, or a sub-system, or a component.

##### Example : Electric Car

Electric motor provides power to move the car

Why do we have an electric motor? }  
or  
What is the purpose of an electric motor? } provide power  
↑  
function

- (ii) FORM : refers to the realization (or HOW?) of a particular function or sub-function.  
realization  $\Rightarrow$  to make real

Example :

How do we provide power → using an  
in an electric car?

Electric motor  
↑  
form

FAST (from GE) is a systematic structured technique to represent relationships between

FUNCTION and FORM  
Why? How?

Definition of the problem: How do we

create a FAST diagram to represent function-form relationships for any system (engineering, biological,....)?



The plan or process for creating a FAST diagram for simple systems is as follows.

1. Understand how the product work
  - playing with the product
  - one source is "howstuffworks.com"
2. Identify the primary or main function of the product, and place it on the extreme right of the diagram
3. Systematically ask "WHY?" and "HOW?" questions, & organize the answers with the
  - "WHYs?" toward the right of the diagram
  - "HOWs?" toward the left " " "
4. Stop when you reach the level of the sub-systems of the product.

Example : Light-Bulb